APPENDIX B

ROTOR CHARACTERISTICS, OPERATING CONDITIONS, AND CALCULATION PARAMETERS

Rotor Physical Description

Articulated blade with flap, lag, and feather hinges at hinge offset e; no precone, prelag, pitch/flap coupling, pitch/lag coupling, flap hinge damper, or structural damping; spar centroid, aerodynamic center, center-of-gravity, and elastic axis at quarter chord.

Geometry	
radius, R	7.62 m (25 ft)
number of blades, N	3
hinge offset, e/R	0.04
linear twist, from center of rotation to tip	$-10°$
chord, c	0.558 m
solidity, σ	0.070
Inertial and structural parameters	
running mass, m	11.97 kg/m
blade weight, M_b	859 N
first mass moment, S_b	320 kg-m
second mass moment, I_b	159 kg-m^2
Lock number, γ	8.43
flatwise bending stiffness, EI_{flat}	8.61×10^4 N-m^2
edgewise bending stiffness, EI_{edge}	1.148×10^6 N-m^2
torsional moment of inertia, I_θ	.004 kg-m^2/m
torsional strength, GJ	8.61×10^4 N-m^2

Three flatwise modes (rigid flapping and two elastic bending modes), two edgewise modes (rigid lag and one elastic bending mode), and one torsion mode (elastic torsion, with a rigid control system) used in the calculation of the blade motion.

Rotating natural frequencies of blade modes (per rev)

Flatwise	Edgewise	Torsion
1.030	0.245	5.836
2.694	3.851	
5.055		

Aerodynamic properties	
section airfoil	NACA 0012
tip loss factor, B	0.97
root cutout	$0.15R$